1.

List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

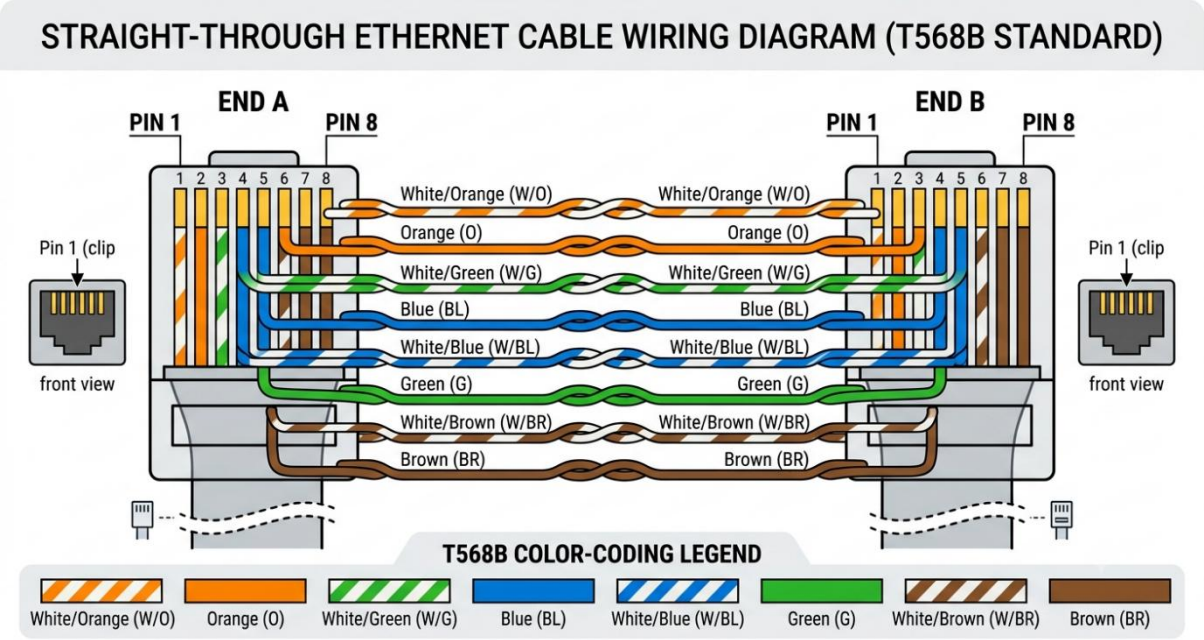
Difference	Ethernet Cable	Fiber Optic Cable
Material	Uses copper wires	Uses glass/plastic fibers
Speed & Distance	Good speed but shorter distance	Very high speed and works over much longer distances
Signal	Uses electrical signals	Uses light signals

Real-world examples

- **Ethernet:** Connecting a **home Wi-Fi router to a laptop/PC**.
- **Fiber Optic:** Connecting buildings in a **college campus backbone or data center**.

2.

Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



3.

Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

Things needed:

- CAT6 cable
- RJ45 connector
- Crimping tool
- Cable cutter/stripper

Steps:

1. Cut the CAT6 cable to the required length.
2. Remove a small part of the outer cover of the cable.
3. Separate and straighten the 8 small wires.
4. Arrange the wires in **T568B order**:
    - 1. White-Orange
    - 2. Orange
    - 3. White-Green
    - 4. Blue
    -

5. White-Blue
   - 6. Green
   - 7. White-Brown
   - 8. Brown
5. Cut the wire ends evenly.
6. Put all 8 wires into the RJ45 connector in the correct order.
7. Insert the connector into the **crimping tool**.
8. Press the tool firmly to lock the connector.
9. Check that all wires are in the correct order.
10. If available, use a **cable tester** to check the cable.

Result: The RJ45 connector is now attached to the CAT6 cable and can be used to connect a computer, router, or switch.

4.

Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences.

Cable Choice for Gaming Café

I would use **Ethernet cables (CAT6)** for the 10 gaming PCs because they are fast, reliable, and affordable.
The PCs are usually placed close to the network switch, so fiber optic is not necessary.
I would use an **RJ45 crimper** to attach the RJ45 connectors to the Ethernet cables.
This setup is easier to install and costs less than using fiber optic cables.